MIL-I-60528B(MU)
27 September 1971
SUPERSEDING
MIL-I-60528A(MU)
12 June 1969

# MILITARY SPECIFICATION
# INITIATOR, CARTRIDGE ACTUATED: DELAY, M30A1 ASSEMBLY

## 1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated: Delay, M30A1, Assembly, Loaded with Cartridge, Delay, M70, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification For |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MS-28741 | - Hose Assembly, Detachable End Fitting, Medium Pressure |

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

| | |
|---|---|
| F8595400 | - Initiator, Cartridge Actuated, Delay, M30A1 Assembly |
| C8798142 | - Box, Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31 or M32A1, with Cartridge, in Fiber Container, M398 |
| C8798143 | - Container, Ammunition, Fiber, M398 for Initiator Cartridge Actuated, Delay, M10 |
| IEL-8595400 | - Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M30A1, Assembly |

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawing and referenced specifications.

3.2 Firing pin release.-The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum.

3.3 Initiator pin security.-The initiator pin of the assembled initiator, less cartridge, cartridge retainer and chamber shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin.

3.4 Breakaway torque.-The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.-The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.



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3.6 Functional.

3.6.1 Ballistics.-Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of size No. 4 aircraft hose, in accordance with MS28741.

TABLE I

| Temperature | Pressure (PSI) | Ignition Delay (See 6.3) |
|---|---|---|
| -65° to -70°F | 500 to 1250 | 2.5 Seconds, Max. |
| 65° to 75°F | 600 to 1500 | 1.80 to 2.20 Seconds |
| 160° to 165°F | 700 to 1700 | 1.6 Seconds, Min. |

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship.-The requirement for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-The component shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.



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4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submittal of product.-The contractor shall submit the product in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

| Lot Size | Sample Size | Major | | Minor | |
|---|---|---|---|---|---|
| | | Accept | Reject | Accept | Reject |
| 1 - 300 | 100% Examination | - | - | - | - |
| 301 - 800 | 50 | 0 | 1 | 1 | 2 |
| 801 - 1800 | 90 | 1 | 2 | 2 | 3 |

4.2.2.1 Cartridge, Delay, M70, Drawing D8593274, covering a detail of Drawing F8595400.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Mixed cartridges | Visual |
| 2. Metal defective | Visual |
| 3. O-ring missing | Visual |
| 4. Sealing disc missing or damaged | Visual |
| 5. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. O-ring split, cut or cracked | Visual |
| 102. Foreign matter | Visual |



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4.2.2.1 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MINOR** | |
| 201. Case scratched, wrinkled, bulged or dented | Visual |
| 202. Varnish missing or inadequate | Visual |

4.2.2.2 Assembly, Drawing F8595400, less Drawings D8593274, D8594860, D8594857 and B8359536.

| | |
|---|---|
| **CRITICAL** | |
| 1. Mixed initiators | Visual |
| 2. Initiator not in cocked position | Visual |
| 3. Safety pin missing or improperly engaged | Visual |
| **MAJOR** | |
| 101. O-ring missing or not lubricated | Visual |
| 102. Thread damaged | Visual |
| 103. Burr | Visual |
| 104. Foreign matter | Visual |
| **MINOR** | |
| None defined | |

4.2.2.3 Assembly, Drawing F8595400, less Drawings D8594857 and B8359536.

| | |
|---|---|
| **CRITICAL** | |
| 1. Cartridge loose | Visual/Manual |
| 2. Cartridge retainer loose | Visual/Manual |
| 3. Metering groove of cartridge retainer missing | Visual |
| 4. Elongated hole of cartridge retainer missing | Visual |
| **MAJOR** | |
| 101. Burr | Visual |
| 102. Foreign matter | Visual |
| **MINOR** | |
| None defined. | |



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4.2.2.4 Assembly, Drawing F8595400, less Drawing B8359536.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Cartridge retainer improperly positioned | Gage |
| 2. Any part missing, except as otherwise classified | Visual |
| 3. Improper assembly | Visual |
| 4. Foreign matter in gas port | Visual |
| **MAJOR** | |
| 101. Length from cap boss face to gas port face, max. | Gage |
| 102. Protrustion of initiator pin, max. | Gage |
| 103. Initiator pin not free to rotate | Manual |
| 104. Safety pin deformed or loose | Visual/Manual |
| 105. Thread damaged or not full | Visual |
| 106. Burr | Visual |
| 107. Foreign matter | Visual |
| **MINOR** | |
| None defined | |

4.2.2.5 Assembly, Drawing F8595400.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Any part missing, except as otherwise classified | Visual/X-Ray 1/ |
| 2. Marking missing, incorrect or illegible | Visual |
| **MAJOR** | |
| 101. Height of cap, max. | Gage |
| 102. Set screw missing, loose, above flush or not covered with barrier material. | Visual |
| **MINOR** | |
| 201. Total length, max. | Gage |
| 202. Location of labels incorrect | Visual |
| 203. Shipping cap missing | Visual |
| 204. Protective finish inadequate or defective | Visual |

1/ The cartridge shall be X-Rayed with the long axis of the cartridge perpendicular to the rays of the machine. The X-Ray negatives, properly identified, shall be forwarded to the procuring activity.



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4.2.2.6 Fiber Container, Drawing C8798143.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Gap between body and cover, max. | Manual |
| 202. Contents loose | Manual |
| 203. Metal ends loose or distorted | Manual |
| 204. Cut or damaged through all impregnated layers | Visual |
| 205. Marking missing, incorrect or illegible | Visual |
| 206. Tape incomplete or badly wrinkled | Visual |
| 207. Length of tear tab, min. | Visual |
| 208. Color of tape incorrect | Visual |

4.2.2.7 Packing Box, Drawing C8798142.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual |
| 202. Hardware missing, loose, broken or improperly engaged | Visual/Manual |
| 203. Strapping missing, loose, broken or improperly assembled | Visual/Manual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of unit incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |



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4.2.3 Inspection testing.-The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.-The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.-The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.-Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the initiator.

4.2.3.4 Residual magnetism.-Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional.-The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics.-Twenty-one units shall be tested to determine compliance with the requirement of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. Failure of any unit to comply with requirements 3.6.1, 3.6.2 and 3.6.3 shall be cause for rejection of the lot.

TABLE III

| Temperature | Sample Size |
|---|---|
| -65° to -70°F | 7 |
| 65° to 75°F | 7 |
| 160° to 165°F | 7 |

4.2.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8595400.



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4.3 Test methods and procedures.

4.3.1 Firing pin release.-The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.-The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.-A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.-The initiator shall be tested for excessive residual magnetism using an approved compass type AN-5766 in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.-The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded the sample units shall be reconditioned in accordance with 4.3.5.1.1.

4.3.5.1.1 Temperature conditioning.-A standard M30A1 Initiator, loaded with a dummy M70 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with drawings C8798142 and C8798143.



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6. NOTES

6.1 *Ordering data.-Procurement documents should specify the title,* number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

6.4 Storage and Shipping Information
Explosive Hazard Classification - Class 1
Storage Compatibility - Group B,E, or N
Department of Transportation (DOT) Hazard - Class C
DOT Marking - "Explosive Power Device - Class C - Handle Carefully - Keep Fire Away".

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